



The Selenium Web Application Testing Framework

Part—1

The more Web applications we use, the more tools are needed to automate the testing process, for which there are many options that simplify the process. Since testing processes and methods vary from one organisation to another, their effectiveness also differs. This article introduces Selenium, a famous framework for testing Web applications, and in particular, its IDE component.

Selenium offers a variety of tools to automate the Web application testing process across multiple platforms. It can run on many browsers and operating systems, and can be controlled by many programming languages, including Python, Perl, PHP, Ruby, Groovy and HTML. Selenium consists of three major tools: Selenium IDE, Selenium Remote Control (RC), and Selenium Grid. In this article, we will cover the first one.

Selenium IDE

Selenium IDE is the open source test automation tool to generate test cases. Firefox supports an add-on for Selenium IDE, with the same name. This is a user-friendly automation tool; anyone can develop and run individual test cases or entire test suites. The Selenium IDE can be used to generate the code for any of the programming languages mentioned above, with just a single click!

Installation

You can install the Firefox add-on from the relevant

section at <http://seleniumhq.org/download/>. Click the link to the .xpi file in Firefox, and it will prompt you to install the add-on. Once the installation succeeds, restart your browser. After that, you can access the add-on via the *Tools -> Selenium IDE* menu. The default screen looks like what is shown in Figure 1.

Recording

Recording is started automatically when the IDE is accessed. You can disable this option; un-check the 'Start recording immediately on open' option in the *Options -> Options -> General* tab.

The recorded actions are stored as *Command*, *Target* and *Value* in the *Table* view. The code can be viewed in the *Source* tab. By default, the code is generated in HTML format. The programming language can be specified via the *Options -> Format* menu.

Verification

During recording, the code is generated for an action. The